



Domain 2: Design for New Solutions



Overview

The second domain of the AWS Certified Solutions Architect Professional exam focuses on designing solutions for specific business objectives. The majority of your work as a Solutions Architect Professional lies in building concrete solutions that follow industry best practices to meet certain regulatory compliances and to serve your customers better. You need to be very knowledgeable about the different AWS services and other external tools that you can use to complement your existing cloud architectures. Having an in-depth understanding of what you're working with will allow you to design a strategy that works best for your customer's business objectives while maintaining the costs in check.

This domain is the biggest one among all other SAP-C02 exam domains, with 29% coverage. Ensure that you know the concept of Infrastructure as code (IaC) and how to deploy your cloud stack across different AWS accounts using AWS CloudFormation. Setting up Continuous Integration/Continuous Delivery (CI/CD) and Change Management processes in AWS Elastic Beanstalk, AWS CodeDeploy, Amazon ECS Anywhere, Amazon EKS Anywhere, and deployment strategies are included. You should also know how to use the Configuration Management tools available in AWS Cloud, such as AWS Systems Manager, AWS Service Catalog, AWS Config, et cetera.

The questions in the actual exam revolve around the following tasks:

- Designing a deployment strategy to meet business requirements
- Designing a solution to ensure business continuity
- Determining security controls based on requirements
- Designing a strategy to meet reliability requirements
- Designing a solution to meet performance objectives.
- Determining a cost optimization strategy to meet solution goals and objectives

In this chapter, we will cover the related topics for designing solutions centered around security, performance, cost optimization and reliability, as well as different deployment strategies in AWS that will likely show up in your Solutions Architect Professional exam.



Using Amazon AppStream 2.0 / Amazon Workspaces for Remote Desktop Operations

Amazon AppStream and Amazon Workspaces are both great services if you need to stream virtual workstations and desktop applications without worrying about hardware. Compared to running virtual machines, virtual workstations can scale and load balance automatically to match user demand. They are made globally available so they can be accessed almost anywhere. Both these services also integrate seamlessly with active directory so you can apply your organization policies and protocols.

When connecting to EC2 virtual machines, you do so via SSH or RDP connection. The method of accessing your Workspaces workstations is different, since you will need a Workspaces client for this. Multiple devices can access the workstation as long as they have the client. During configuration, Workspaces offers multiple bundles as a base, which include a mix of hardware and software options for you to choose from. With regards to overall expenses, larger corporations benefit more from Amazon Workspaces due to the service requiring you to use an active directory for managing your users.

If you only need to share access to your applications and not a whole desktop environment then Amazon Appstream is the service to use. You can stream your applications to any computer without having to provision and operate hardware. All you need is a HTML5-capable browser. Appstream allows you to restrict the applications that are available for use to your users and prevent them from using any other unrelated applications. Appstream fleets can be linked to an active directory, but this is not mandatory. Appstream also allows you to share your applications on a massive scale, which is normally constrained by hardware, thanks to the capacity made available by AWS infrastructure.

References:

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/desktop-app-streaming-services.html>

<https://aws.amazon.com/workspaces/features/?nc=sn&loc=2>

<https://aws.amazon.com/appstream2/features/?nc=sn&loc=3>

[AWS re:Invent 2018: Desktops & Applications to AWS with Amazon WorkSpaces & AppStream 2.0](#)

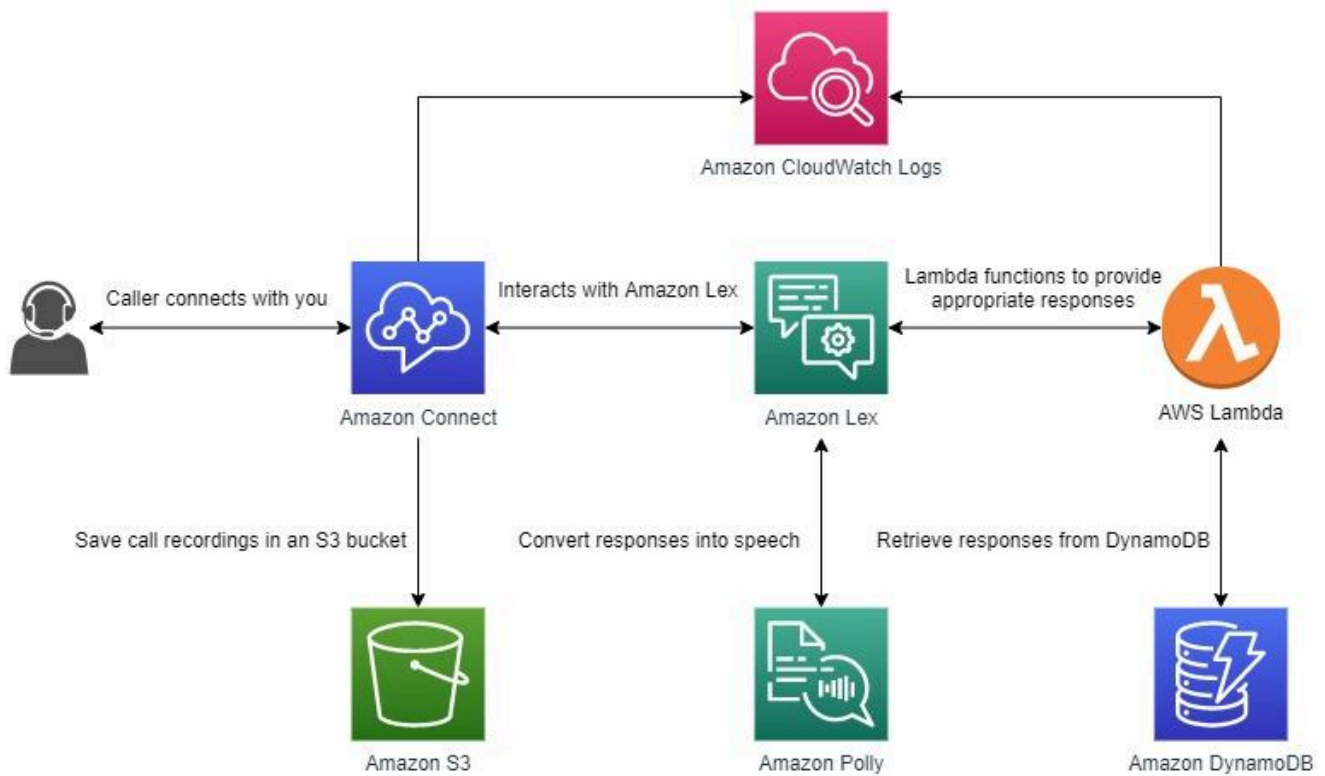


Using Amazon Connect, Amazon Lex, and Amazon Polly For Chat and Call Functionality

If your business involves taking calls and communicating with customers via phone, you can quickly set up a contact center using Amazon Connect, Amazon Lex, and Amazon Polly. Amazon Connect is an easy to use omnichannel cloud contact center that supports many useful features for voice and chat operations. Customers can easily call into your Amazon Connect contact center using any phone and speak to an agent. You can also set up web and voice chat via APIs. You can design workflows that route calls and messages to the appropriate responder. If you need to perform language processing when you receive calls, Amazon Connect can be integrated with Amazon Lex for *Natural Language Understanding* and automated customer interactions. For text-to-speech messages, you may also use Amazon Polly. Calls can also be recorded and stored in Amazon S3 for future analytics. Lastly, it is common nowadays to have interactive chat boxes running on your website for visitors. You can easily configure your automated responses for some expected queries and have them invoke Lambda functions to handle any external processing.

Amazon Connect, Amazon Lex, and Amazon Polly make it very convenient for call centers and businesses that offer phone and chat support to quickly set up a production-ready contact center. These services allow you to respond back quickly to your customers and offer top quality support. Amazon Connect provides out-of-the-box integrations for Salesforce and Zendesk if you are using these tools already. You can also run machine learning workloads on your recordings which will further improve your engagement with your customers. If you have a chatbot running on your website, the recording data will definitely serve useful in training your bot.

Amazon Lex is a machine learning service that you can use to develop client-facing chatbots for your customers. You can launch either a Text-based or Voice-based chatbot with Amazon Lex and then integrate it on your website or to the other AWS services. This minimizes the dependency of having a virtual agent for your conversational Interactive Voice Response (IVR) system, corporate website, or other customer-facing system. It can also lower down the costs of companies in maintaining its contact center.



References:

<https://aws.amazon.com/blogs/aws/amazon-connect-customer-contact-center-in-the-cloud/>
<https://aws.amazon.com/blogs/aws/new-omnichannel-contact-center-web-and-mobile-chat-for-amazon-connect/>



Using Amazon WorkDocs for Secure Document Management and Collaboration

Most of the users in AWS use more well-known storage services such as Amazon S3, Amazon EBS, and Amazon EFS for file storage and file sharing purposes. Others may use third-party tools such as Google Drive, DropBox, and Microsoft OneDrive, which may offer more functionality for the user, or even local storage drives in case there are compliance requirements to meet. While it is perfectly fine to use these services, don't forget that AWS also offers its own product for secure content storage and collaboration through Amazon Workdocs. Compared to storage services such as Amazon S3 or your own storage devices, Amazon WorkDocs provides the following additional benefits:

- 1) Amazon WorkDocs offers unlimited versioning. A new version of a file is created every time you save it.
- 2) You can invite other AWS users to view, contribute to, or co-own your files by entering user names, group names, and email addresses. You can also request specific feedback with a personal message and set a deadline.
- 3) Amazon WorkDocs lets you use your Active Directory to manage your users.
- 4) Amazon WorkDocs lets you control who can access, comment, and download or print your files. You can "lock" files to ensure that edits are not overwritten by other contributors, eliminating the need to coordinate changes.
- 5) You can create an approval workflow to route documents and other files stored in WorkDocs to one or more users for their approval. This allows you to track and manage your documents easier.
- 6) *Amazon WorkDocs Drive* is a desktop application that connects your computer to your Workdocs filesystem so that all of your files are available on-demand from your device, eliminating the need to use network shares.
- 7) You can use AWS API to interact with your WorkDocs filesystem programmatically.

Solutions Architect Professional Exam Notes:

So when should I choose Amazon WorkDocs over other options?

Since you will be taking an AWS exam, third-party options are already out of the question. But when you are made to choose between the different AWS Storage Services and Amazon WorkDocs, keep an eye out for key terms such as "content management", "collaboration" and "document sharing". These usually indicate that the scenario requires a convenient document collaboration tool which is what Amazon WorkDocs is. Lastly, be sure to read through the benefits listed above, since they will be useful in letting you evaluate if Amazon WorkDocs is the better choice.

Reference:

<https://aws.amazon.com/workdocs/>

Using AWS Data Exchange for Proprietary Data Access in Amazon Redshift

AWS Data Exchange allows data providers to share data products with subscribers, who can then use this data for their own purposes, such as training a machine learning model. For businesses that process and store proprietary data in Amazon Redshift, the challenge often lies in securely and sharing this data with external customers. If you want to monetize your data, making them available to users and managing the access takes a lot of work.

AWS Data Exchange simplifies this process significantly. To share data processed within Amazon Redshift, the steps are as follows:

1. Create a Datashare within Amazon Redshift: As a data provider, you start by creating a datashare in your Amazon Redshift cluster. This involves selecting the specific schemas, tables, views, and user-defined functions that you intend to share. The datashare acts as a container for the data you wish to monetize, allowing you to control exactly what is shared and ensuring that your proprietary data is handled securely.
2. Import the Datashare into AWS Data Exchange: Once the datashare is created, the next step is to import it into AWS Data Exchange. This process involves creating a new dataset within AWS Data Exchange, into which your Redshift datashare is incorporated. This dataset forms the basis of the product you will offer to potential subscribers.
3. Create and Publish Your Product: With your dataset in place, you can now create a product in AWS Data Exchange. This product encapsulates the data you wish to share and can include additional details such as pricing, terms of use, and a description of the data. Once your product is ready, you publish it on AWS Data Exchange, making it available to potential subscribers.
4. Subscription by Customers: After publication, customers can subscribe to your product. Subscribers are granted read-only access to the data you have shared, allowing them to view this data within their without the ability to modify the original data.

References:

<https://docs.aws.amazon.com/data-exchange/>

<https://aws.amazon.com/blogs/aws/new-aws-data-exchange-for-amazon-redshift/>

Implementing DDoS Resiliency in AWS

Websites and web applications are always under threat of security attacks. Today, there are many ways for people to exploit a vulnerability to steal data, cause huge amounts of downtime, and prevent your applications from recovering swiftly and properly. One infamous security attack that affects thousands of websites and



endpoints each day is known as Denial of Service. A Denial of Service (DoS) attack is a deliberate attempt to make your website or application unavailable to users, by flooding it with network traffic for example. To achieve this, attackers use a variety of techniques that consume large amounts of network bandwidth or tie up other system resources, disrupting access for legitimate users.

A more extreme version of this is DDoS, or distributed denial of service, where the attacker uses multiple computers to perform the attack. DDoS attacks are most common at layers 3, 4, 6, and 7 of the OSI model. To address this, we will be taking a look at some of the tools that AWS provides to you as their customer to defend against these types of attacks.

AWS WAF is a web application firewall that helps protect your web applications or APIs against common web exploits such as flood attacks, XSS, and SQL injection attacks. In AWS WAF, you can use a web access control list to protect your AWS resources. You create a web ACL and define its protection strategy by adding rules. Rules define criteria for inspecting web requests and specify how to handle requests that match the criteria. You set a default action for the web ACL that indicates whether to block or allow those requests that pass the rules inspections. To help you get started, AWS and AWS Marketplace vendors offer preconfigured WAF rule groups. These rule groups contain a set of rules that will help protect you from common security threats and exploits. The resource types that you can protect using WAF web ACLs are Amazon CloudFront distributions, Amazon API Gateway REST APIs, and Application Load Balancers.

To mitigate a potential layer 7 DDoS attack, create conditions in AWS WAF that match the unusual behavior. Configure the web ACL to count the requests that match the rules. If the volume of requests continues to be unusually high, change your web ACL to block those requests.

AWS Shield is a service that protects your resources from DDoS attacks. AWS Shield Standard provides protection against common and most frequently occurring infrastructure (layer 3 and 4) attacks like SYN/UDP floods, reflection attacks, and others to support high availability of your applications on AWS. AWS Shield Standard is automatically included in every AWS account at no extra cost. AWS Shield Advanced is an optional paid subscription that provides enhanced protections for your applications running on EC2 with EIPs, ELB load balancers, CloudFront distributions, AWS Global Accelerator, and Route 53 resources against more sophisticated and larger attacks. Using AWS Shield Advanced with EIPs allows you to protect Network Load Balancer (NLBs). If you are using Amazon CloudFront and Amazon Route 53, these services receive comprehensive availability protection against all known infrastructure attacks.

Amazon Route 53 and Amazon CloudFront are two services that you should employ to maximize your defense against DoS attacks in AWS. Evident from the previous security services, Route 53, and CloudFront both easily integrate with AWS WAF and Shield. AWS edge locations also provide an additional layer of network infrastructure that provides these benefits to any web application that uses CloudFront and Route 53.



Persistent connections and variable time-to-live (TTL) settings can be used to offload traffic from your origin, even if you are not serving cacheable content. These features mean that using CloudFront reduces the number of requests and TCP connections back to your origin which helps protect your web application from HTTP floods. Amazon CloudFront only accepts well-formed connections, which helps prevent many common DDoS attacks, like SYN floods and UDP reflection attacks, from reaching your origin. DDoS attacks are also geographically isolated close to the source which prevents the traffic from impacting other locations.

Amazon Route 53 is a highly available and scalable domain name system. It uses shuffle sharding and anycast striping to prevent a DDoS attack from affecting your website's availability. With shuffle sharding, each name server in your delegation set corresponds to a unique set of edge locations and internet paths. If one of these name servers becomes unavailable, users can retry their request and receive a response from another name server at a different edge location. Anycast striping allows each DNS request to be served by the most optimal location, spreading the network load and reducing DNS latency. Additionally, Route 53 can detect anomalies in the source and volume of DNS queries, and prioritize requests from users that are known to be reliable.

i Solutions Architect Professional Exam Notes:

In the exam, it is common to see AWS WAF being used in conjunction with CloudFront or ALB to defend against security attacks. Remember that WAF does not support integration with NLB or CLB, nor does it support direct EC2 integration. You can use WAF with Amazon ECS, as long as your ECS cluster has an ALB in front. Also, take note that AWS WAF is not the best service to use against DDoS attacks, due to the fact that a lot of the mitigation effort requires manual activity. If you are given the option to select AWS Shield in a DDoS scenario, you should lean more on this option since the mitigation effort is automated.

Other options you might encounter include using load balancers and auto scaling groups, or enforcing rules in security groups and network ACLs. Although these services can help prolong your website's availability and keep a few attacker IP addresses at bay, they do not protect your web servers completely from the different security threats. You won't be able to filter out all of the IP addresses if it is a large scale attack, and your instances can only scale out so much until you hit your max size.

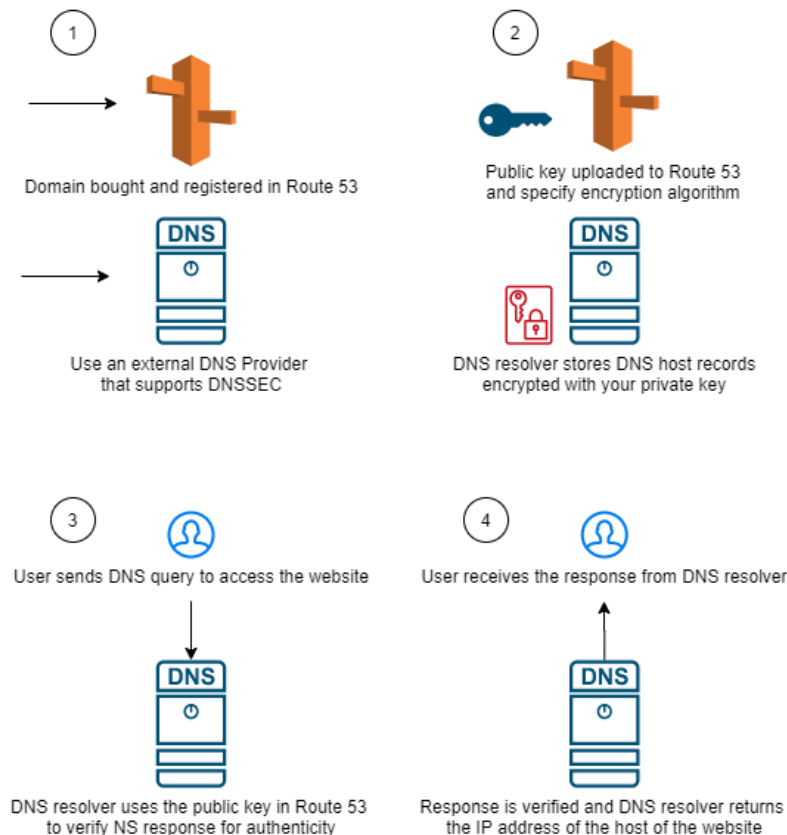
References:

https://d1.awsstatic.com/whitepapers/Security/DDoS_White_Paper.pdf
<https://docs.aws.amazon.com/waf/latest/developerguide/waf-chapter.html>
<https://docs.aws.amazon.com/waf/latest/developerguide/ddos-overview.html>

Configuring DNSSEC for a Domain in Route 53

Sometimes, dodgy people just feel like doing dodgy stuff. It is up to you to keep yourself and your resources protected from their maliciousness. Everything you expose on the Internet needs to be properly secured. And yes, that includes your own web domain. Attackers sometimes hijack traffic to Internet endpoints such as web servers by forging DNS data and fooling DNS resolvers to route users to the IP addresses provided by the attackers, for example, to fake websites. This type of attack is known as DNS spoofing or a man-in-the-middle attack. Not only does this mean that you lose reputation for your websites, but also the fact that your other Internet endpoints might be affected as well.

You can protect your domain from this type of attacks by configuring **Domain Name System Security Extensions (DNSSEC)**, a protocol for securing DNS traffic. DNSSEC works by using asymmetric encryption to secure your DNS records and verify domain requests sent to the DNS resolver. Amazon Route 53 supports DNSSEC for domain registration as well as DNSSEC for its DNS service.



Reference:

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/domain-configure-dnssec.html>



Configuration Management in AWS with AWS Systems Manager

When managing servers, the traditional approach for applying changes or performing checks involves manually logging into each server to update configurations or run diagnostics. This method is manageable for a single server or a small setup. However, at scale, with potentially hundreds or thousands of servers spread across different environments, this approach quickly becomes impractical. It's not only time-consuming but also increases the risk of inconsistencies and errors.

Imagine you're managing hundreds of servers and one employee at your work accidentally misconfigures a server, perhaps by mistakenly altering firewall settings. This single misstep could expose your organization to significant security risks, potentially leading to data breaches.

AWS Systems Manager State Manager for Consistency

Systems Manager State Manager allows you to define a desired configuration state for your servers, covering everything from security policies and software installations to system settings. Once defined, State Manager continually enforces these configurations across your entire server fleet. For example, you could use State Manager to ensure that firewall rules across all servers only permit traffic from trusted IP addresses and block all unauthorized access attempts. If there are changes to these settings on any server, State Manager would automatically detect this deviation from the desired state and revert the settings back to the correct configuration.

Responding to Emergencies with Run Command

For those times when urgent interventions are required—be it deploying emergency patches or executing critical commands across multiple servers, Systems Manager Run Command is helpful. It allows you to execute commands on your servers at scale, without the need to log into each one individually. This capability is crucial for quickly responding to vulnerabilities or ensuring that all servers are running the latest software, providing a layer of agility and security that is indispensable in modern server management.

References:

<https://docs.aws.amazon.com/systems-manager/latest/userguide/systems-manager-state.html>
<https://docs.aws.amazon.com/systems-manager/latest/userguide/run-command.html>



Using Lambda@Edge for Low Latency Access to your Applications

Perhaps you have products that are catering to a global audience, or your traffic is being served worldwide via the vast network channel of Amazon CloudFront. Although you are able to serve your content very quickly thanks to the CDN service, this will only be useful if all the content you serve is cacheable content. If perhaps you have an application that runs in one region while your customers are in other locations then surely they will experience the latency in the responsiveness of your app. Running multiple copies of your infrastructure and applications in different regions can be a solution to this problem, but it is costly and has a lot of management overhead. A better option would be to actually bring your applications closer to your customers using the multiple Edge Locations placed around the globe using Lambda@Edge.

Re-architecting can be scary for some people, since it involves a lot of money, effort and time to successfully migrate a working application to serverless. Though in many cases, re-architecting actually gives a lot of advantages in return. AWS Lambda functions specifically offer a rich set of features and integrations that you can use to improve the performance of your web applications, such as Lambda@Edge with Amazon CloudFront. With Lambda@Edge, your Node JS and Python Lambda functions are executed nearest to your customer's location, and it easily scales as well to keep up with the demand. This significantly reduces latency and improves the user experience.

When you associate a CloudFront distribution with a Lambda@Edge function, CloudFront intercepts requests and responses at CloudFront edge locations. You can execute Lambda functions when the following CloudFront events occur:

- When CloudFront receives a request from a viewer (viewer request)
- Before CloudFront forwards a request to the origin (origin request)
- When CloudFront receives a response from the origin (origin response)
- Before CloudFront returns the response to the viewer (viewer response)

Steps to deploy a Lambda@Edge function:

1. To deploy your Lambda functions at CloudFront Edge Locations, you must first create and publish a Lambda function in the US-East-1 (N. Virginia) Region. Remember that the language should be either Node JS or Python.
2. After publishing it, choose (or create) the CloudFront distribution to be associated with and modify the cache behavior.
3. Select the event type or the *trigger* for your function. After the trigger is created, your Lambda function is now replicated around the world.
4. Verify that your function runs properly. If you receive an error saying that CloudFront cannot execute your Lambda function, be sure to add a policy that allows this action in the IAM role of your function. You can also consult CloudWatch Logs for further information.

- To update your function, you must edit the \$LATEST version of the function in the US-East-1 (N. Virginia) Region. Then, before you set it up to work with CloudFront, you publish a new numbered version.

Step 1: Select delivery method
Step 2: Create distribution

Lambda Function Associations

CloudFront Event: **Select Event Type** (dropdown menu open showing: Select Event Type, Viewer Request, Viewer Response, Origin Request, Origin Response)

Lambda Function ARN:

Include Body: ☐

Enable Real-time Logs: ☐

Distribution Settings

Price Class: **Use All Edge Locations (Best Performance)** (dropdown menu)

Figure: Configuring your CloudFront to add a Lambda function

Deploy to Lambda@Edge

Configure CloudFront trigger

Distribution
The CloudFront distribution that will send events to your Lambda function.

Cache behavior
Choose the cache behavior you would like this Lambda function to be associated with.

CloudFront event
Choose one CloudFront event to listen for.

☐ Include body
Select "Include body" if you want to read the request body for viewer request or origin request events.
[Learn more.](#)

Confirm deploy to Lambda@Edge

☐ I acknowledge that on deploy a new version of this function will be published with the above trigger and replicated across all available AWS regions.

Lambda will add the necessary permissions for Amazon CloudFront to invoke your Lambda function from this trigger.
[Learn more](#) about the Lambda permissions model.

Cancel **Deploy**

Figure: Adding a trigger on your Lambda function and specifying the CloudFront distribution



Example use cases of Lambda@Edge include:

- Work as an extension of or replacement for your origin. This enables you to do everything from simple HTTP request and response processing at the edge to more advanced functionalities, such as website security, real-time image transformation, intelligent bot mitigation, search engine optimization, and more.
- Adding HTTP security headers on all origin responses without having to modify your application code on your origin. This helps improve security and privacy for your users and content providers, while using CloudFront to deliver the content at low latencies.
- Working with Amazon Cognito to provide user authentication for your applications based on location. You can also filter out unauthorized requests before they reach your origin infrastructure.

References:

<https://aws.amazon.com/blogs/networking-and-content-delivery/reducing-latency-and-shifting-compute-to-the-edge-with-lambdaedge/>
<https://aws.amazon.com/lambda/edge/>
<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/lambda-at-the-edge.html>

Setting Up an ELK (ElasticSearch, Logstash and Kibana) Stack Using Amazon OpenSearch

So what is an ELK stack, and what makes it popular in log analytics? ELK is an acronym for three popular open-source projects — ElasticSearch, Logstash, and Kibana. The ELK stack allows you to aggregate logs from all your systems and applications (logstash), analyze these logs (elasticsearch), and create visualizations for application and infrastructure monitoring, faster troubleshooting, security analytics, and more (kibana). You can imagine that in a public cloud space where there are lots of applications and services producing large amounts of log data, one will need a robust and scalable solution to manage these logs and obtain value-adding information.

	Elasticsearch	Logstash	Kibana
Definition	An open-source, RESTful, distributed search and analytics engine built on Apache Lucene.	An open-source data ingestion tool that allows you to collect data from a variety of sources, transform it, and send it to your desired destination.	An open-source data visualization and exploration tool for reviewing logs and events.
How it works	You forward data in the form of JSON documents to Elasticsearch using the API or ingestion tools such	Logstash ingests data from multiple sources, then the data is transformed through a series of filters	Logs and documents sent to Elasticsearch can be visualized in Kibana for graphical views and



	as Logstash and Amazon Kinesis Firehose. Elasticsearch automatically stores the original document and adds a searchable reference to the document in the cluster's index. You can then search and retrieve the document using the Elasticsearch API.	which you design, and finally outputs your data into a "stash".	aggregated representations of your data.
Use cases	Log search, document indexing, log storage, and more.	Data transformation to make the output usable by your receiving applications.	Log and time-series analytics, application monitoring, and operational intelligence use cases.

So why use Amazon OpenSearch Service if you can deploy your own ELK stack? Amazon OpenSearch Service is a fully managed service, cost-effective, and can run at petabyte-scale. You can deploy your own stack onto an EC2 instance or your own servers for that matter, but you do get that additional management overhead as well as configuring scaling to meet demand. Amazon OpenSearch already offers support for Elasticsearch APIs, built-in Kibana, and integration with Logstash, so transitioning between an AWS and a non-AWS ELK deployment is very easy to do. Lastly, Amazon OpenSearch integrates with other AWS services such as Amazon Data Firehose, Amazon CloudWatch Logs, and AWS IoT, giving you the flexibility to select the data ingestion tool that meets your use case requirements.

i Solutions Architect Professional Exam Notes:

Whenever you encounter Elasticsearch in AWS, always consider the options that discuss the Amazon OpenSearch service. It is the most convenient, scalable, and cost-effective solution that you can use to run your ELK stack on AWS. If you need to migrate an on-premises ELK stack to AWS, deploy your Amazon OpenSearch domain first and configure your new cluster. After that, you can utilize AWS Database Migration Service (AWS DMS) to migrate data to Amazon OpenSearch Service from all AWS DMS-supported sources, which currently are:

- Oracle DB
- MS SQL Server
- MySQL
- MariaDB
- PostgreSQL
- MongoDB

- SAP ASE
- IBM Db2
- Azure SQL Database
- Amazon Aurora
- Amazon S3

If your log source is not in this list, you can instead pause the ingestion and export your data and indexes from your existing Elasticsearch. Then import it into your new OpenSearch cluster. Once done, you can repoint your log ingestion to use your new cluster. The open-source Elasticsearch is a great tool to use, but it still involves some additional learning from the user. So unless the scenario calls for Elasticsearch explicitly, consider your other options for log management first such as Cloudwatch Logs or S3.

Reference:

<https://aws.amazon.com/elasticsearch-service/the-elk-stack/>

<https://aws.amazon.com/blogs/database/introducing-amazon-elasticsearch-service-as-a-target-in-aws-database-migration-service/>

Data Analytics and Visualization Using Amazon Athena and Amazon Quick

Performing data analytics in AWS has never been easier thanks to the wide array of services at your disposal. With Amazon S3, you can cost-effectively build and scale a data lake of any size in a secure environment where data is durably stored. Amazon S3 is an object storage solution so it supports almost all kinds of file types. Once you have built your own data lake, you can use the different services that readily integrate with Amazon S3, such as Amazon Athena, to perform data analytics and data processing.

Amazon Athena is a service that lets you run queries on your S3 objects using SQL. If the files stored in your bucket have a common format (let's say load balancer logs for example), you can create a table in Amazon Athena pointing to your S3 bucket and define the schema used by your files. Once the table is generated, you can run your standard SQL queries and Amazon Athena will handle the parsing based on the schema you defined. The results of your queries are saved in a different S3 bucket in case you need them later on. Below is an example of creating a table to parse application load balancer logs:

```
CREATE EXTERNAL TABLE IF NOT EXISTS myalblogs (  
  type string,  
  time string,  
  elb string,  
  client_ip string,  
  client_port int,  
  target_ip string,  
  target_port int,  
  request_processing_time double,  
  target_processing_time double,  
  response_processing_time double,  
  elb_status_code string,
```



```
target_status_code string,
received_bytes bigint,
sent_bytes bigint,
request_verb string,
request_url string,
request_proto string,
user_agent string,
ssl_cipher string,
ssl_protocol string,
target_group_arn string,
trace_id string,
domain_name string,
chosen_cert_arn string,
)
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.RegexSerDe'
WITH SERDEPROPERTIES (
  'serialization.format' = '1',
  'input.regex' = '([^\ ]*) ([^\ ]*) ([^\ ]*) ([^\ ]*):([0-9]*) ([^\ ]*):-([0-9]*) ([-\.0-9]*) ([-\.0-9]*)
([-\.0-9]*) ([-\.0-9]*) (-|[-0-9]*) ([-0-9]*) ([-0-9]*) \"([^\ ]*) ([^\ ]*) (-|([^\ ]*)\" \"([^\"]*)\" \"
([A-Z0-9-]+) ([A-Za-z0-9-.-]*) ([^\ ]*) \"([^\"]*)\" \"([^\"]*)\" \"([^\"]*)\" ([-\.0-9]*) ([^\ ]*)
\"([^\"]*)\" \"([^\"]*)\" \"([^\ ]*)\" \"([^\s]+?)\" \"([^\s]+?)\" \"([^\ ]*)\" \"([^\ ]*)\"')
LOCATION 's3://alb-logbucket/AWSLogs/1234567890/elasticloadbalancing/us-east-1/';
```

Amazon Athena also supports Amazon Quick for interactive data visualization of your query results. Before you try to read files from S3 buckets, make sure that you grant Amazon Quick access to them. You also need to grant access to Amazon Athena to run queries.

To start visualizing the data, follow the steps below:

- 1) Go to the homepage of Amazon Quick and choose *Manage Data*.
- 2) Create a new data set.
- 3) For the data source, choose Athena and fill in additional details such as a data source name.
- 4) On the *Choose your table* screen, you can write your own SQL script or choose an existing database and table that you have created in Athena.
- 5) On the Finish data set creation page, choose how you want Amazon Quick to handle your data.
 - a) You can load your data into memory with *Importing Data into SPICE*
 - b) You can query your data directly without using SPICE. With this option, you rerun the query each time you open the analysis or dashboard.
- 6) Select *Visualize* to create the dataset and analyze your data.

References:

<https://aws.amazon.com/products/storage/data-lake-storage/>

<https://docs.aws.amazon.com/whitepapers/latest/big-data-analytics-options/amazon-athena.html>

<https://aws.amazon.com/quick/>

Using AWS Transfer Family for FTP Use Cases

Multiple industries rely on secure channels to transfer their data back and forth between different servers and storages in AWS. Some might also have compliance requirements to follow, which often requires strong encryption for data in-transit. Having to manage your own servers and secure channels to transfer sensitive files to AWS can add unnecessary overhead to your IT team. So instead, the AWS Transfer Family is a set of solutions that removes the operational overhead so your team can focus on your file transfers.

The AWS Transfer Family is the aggregated name of **AWS Transfer for SFTP (Secure Shell File Transfer Protocol)**, **AWS Transfer for FTPS (File Transfer Protocol over SSL)**, and **AWS Transfer for FTP**. The AWS Transfer Family offers fully managed support for the transfer of files over SFTP, FTPS, and FTP directly into and out of Amazon S3. You can seamlessly migrate your file transfer workflows by maintaining existing client-side configurations for authentication, access, and firewalls, so no changes need to be made for your customers, partners, and internal teams, or their applications. Data stored in Amazon S3 can be processed for multiple types of workloads, and can be moved around in your internal AWS network securely.

FTP	FTPS	SFTP
FTP is a network protocol used for the transfer of data. FTP uses a separate channel for control and data transfers. FTP uses cleartext and does not support encryption of traffic, which is also why the server does not allow you to use FTP over public networks. If traffic needs to traverse the public network, secure protocols such as SFTP or FTPS should be used.	FTPS is an extension of FTP that uses Transport Layer Security (TLS) and Secure Sockets Layer (SSL) cryptographic protocols to encrypt traffic. FTPS allows encryption of both the control and data channel connections either concurrently or independently.	SFTP is a network protocol used for secure transfer of data over the Internet. The protocol supports the full security and authentication functionality of SSH, such as the use of SSH keys.

The AWS Transfer Family provides you with a fully managed, highly available file transfer service with auto-scaling capabilities. Your end users' workflows remain the same, while data uploaded and downloaded over the different FTP protocols are stored in your S3 bucket. You set up your users by integrating an existing identity provider like Microsoft AD or LDAP for authentication. You should also assign IAM Roles to your users to provide access to your S3 buckets. A VPC is required to host FTP server endpoints.

Reference:

<https://aws.amazon.com/aws-transfer-family/>



A Single Interface for Querying Multiple Data Sources with AWS AppSync

Suppose you have multiple data sources that use different APIs to communicate with your applications. Your infrastructure might turn out to be very complex which can make it difficult for your users to fetch the data from all these sources, especially if they need it in real time. With AWS AppSync, you can simplify the process by having a single interface that users can interact with, and the interface will take care of fetching the data from multiple sources for you through the help of GraphQL technology.

For those who are unfamiliar with GraphQL, it is a data query and manipulation language that enables client apps to fetch, change, and subscribe to data from servers. The client specifies exactly what data it needs, and GraphQL aggregates the data from multiple sources and returns it to the client in JSON format. GraphQL also includes a feature called “*introspection*” which lets new developers on a project discover the data available without requiring knowledge of the backend.

The following are the features of AWS AppSync:

- Real-time data access and updates through subscriptions. When there are changes in the data, the results can be passed down to subscribed clients immediately using either MQTT over WebSockets or pure WebSockets.
- Offline data synchronization with Amplify DataStore that provides a queryable on-device datastore for web, mobile, and IoT developers.
- Data querying, filtering, and search in apps. AWS AppSync supports AWS Lambda, Amazon Aurora Serverless, Amazon DynamoDB, Amazon Elasticsearch, and HTTP endpoints as data sources.
- Server-side data caching capabilities. It reduces the need to directly access data sources all the time. Frequently accessed data are stored in high speed in-memory managed caches, and delivered at low latency.
- Several levels of data access management and authorization through AWS IAM Roles, integration with Amazon Cognito User Pools for email and password functionality, social identity providers (Facebook, Google+, and Login with Amazon), and enterprise federation with SAML.

Use Cases of AWS AppSync include:

- Create dashboards and web and mobile applications that need collective real-time data from multiple sources.
- Access and combine data from microservices running in containers in a VPC, behind a REST API endpoint, a GraphQL API endpoint, and more in a single interface in AppSync.
- Retrieve or modify data from multiple data sources (SQL, NoSQL, search data, REST endpoints, and serverless backends) with a single query.
- Automatically synchronize data between mobile/web apps and the cloud with AWS AppSync and AWS Amplify DataStore.



AWS AppSync SDKs support iOS, Android, and JavaScript, and span web frameworks such as React and Angular as well as React Native and Ionic. You can also use open source clients to connect to the AppSync GraphQL endpoint such as generic HTTP libraries or simple CURL commands.

Solutions Architect Professional Exam Tips:

If you look at it, AWS AppSync sounds awfully similar to Amazon API Gateway. While they do provide API functionalities for your applications, they differ in the kind of APIs provided. AppSync has GraphQL while API Gateway has RESTful and WebSocket APIs. There are also numerous other distinctions such as throttling features, integration features, request validation and custom response features, latency requirements, security features, etc. So do be careful in reading your exam scenario so you can discern what is the best solution for the item.

References:

<https://aws.amazon.com/appsync/>

<https://aws.amazon.com/blogs/mobile/appsync-microservices/>

[Data Driven Applications with AWS AppSync and GraphQL](#)